

Paper Practice 1.3 • WORKED KEY**Properties of Real Numbers**

Show all of your work in the space provided. When a problem asks for a property, write its FULL name. Circle your final answers.

- 1.** Name the property of equality shown by each statement. (Properties of Equality)

a. If $14 + 6 = 20$, then $20 = 14 + 6$.

Answer: Symmetric

b. $n + 9 = n + 9$

Answer: Reflexive

c. If $x = 7$ and $7 = 3 + 4$, then $x = 3 + 4$.

Answer: Transitive

- 2.** Name the property shown by each statement. (Number Properties)

a. $23 \cdot 1 = 23$

Answer: Multiplicative Identity

b. $8 + (-8) = 0$

Answer: Additive Inverse

c. $(2/5)(5/2) = 1$

Answer: Multiplicative Inverse

d. $17 \cdot 0 = 0$

Answer: Multiplicative Property of Zero

- 3.** Evaluate: $6 - 2^2 + 9(1)$ (Evaluate)

Answer: $6 - 4 + 9 = 11$

- 4.** Evaluate: $(3/4)(4/3) \cdot 12$ (Evaluate)

Answer: $(3/4)(4/3) = 1$, so the product is 12
(Multiplicative Inverse)

- 5.** Evaluate: $(18 - 2 \cdot 3 + 5) \cdot 0$ (Evaluate)

Answer: 0 – by the Multiplicative Property of Zero
you never have to simplify inside

- 6.** Use the commutative and associative properties to evaluate $25 + 38 + 75 + 62$.
Show the regrouping. (Mental Math)

Answer: $(25 + 75) + (38 + 62) = 100 + 100 = 200$

7. Use the commutative and associative properties to evaluate $4 \cdot 17 \cdot 25$. Show the regrouping. (Mental Math)

Answer: $(4 \cdot 25) \cdot 17 = 100 \cdot 17 = 1700$

8. Use the properties to evaluate $36 + 59 + 64 + 41$. Show the regrouping. (Mental Math)

Answer: $(36 + 64) + (59 + 41) = 100 + 100 = 200$

9. Jade spends \$41, \$36, \$39, and \$44 on groceries over four weeks. Use one of the properties to find her total, and name the property you used. (Apply)

Answer: $(41 + 39) + (36 + 44) = 80 + 80 = \160 .
Commutative and Associative Properties of Addition.

10. Name the property used in each step. (Justify)

$$\begin{aligned} 12 + 45 + 8 + 55 \\ = 12 + 8 + 45 + 55 \end{aligned}$$

Answer: Commutative Property of Addition – the ORDER changed

$$= (12 + 8) + (45 + 55)$$

Answer: Associative Property of Addition – the GROUPING changed

$$= 20 + 100 = 120$$

Answer: Substitution (Simplify)

11. Error Analysis – Gus said that $(5 + 3) + 7 = 5 + (3 + 7)$ shows the Commutative Property. Identify his mistake. (Error Analysis)

Answer: The numbers stayed in the same ORDER – only the parentheses moved. That is the ASSOCIATIVE Property. Commutative changes order; associative changes grouping.

12. Write your own example of the Multiplicative Inverse Property, then explain in one sentence why the product is always 1. (Explain)

Answer: Sample: $(7/9)(9/7) = 1$. Multiplying a fraction by its reciprocal puts the same factors in the numerator and denominator, so everything divides out.